

Financial Mathematics

(MATH35400/MATHM5400)

Problem Class 2 – 30th January 2026 ¹

Problems:

1. **(2021 exam, Q1)** Stocks of telecommunication company *PlinTo* are currently traded at $S_0 = £8$ per share. A financial company wants to price a European call option on *PlinTo* shares with strike price $K = £5$ maturing in exactly one year. They have access to a bank account with an annual interest rate $r = 0.125$.
 - (a) Alex suggests to model the stock by assuming that the stock price in a year's time S_1 is either $£11$ with probability 0.6 or $£3$ with probability 0.4. Alex works out the expected value of the call option as 3.6 and wants to use that as the time 0 price for the option.
 - i. Find a replicating portfolio for the European call option, determine its time 0 value C_0 and compare with the price that Alex has calculated.
 - (b) Kim has doubts about Alex' model and is concerned that the stock price might increase to $S_1 = £20$.
 - i. Suppose the company sells one European call option for C_0 from (a) and builds up the replicating portfolio from (a) at time 0. What would be the loss at time 1 if the stock price would actually increase to $£20$?
 - ii. Kim suggests to consider a model with three states $\omega_i, i = 1, 2, 3$ where $S_1(\omega_1) = £20$, $S_1(\omega_2) = £11$ and $S_1(\omega_3) = £3$ and where $\mathbb{P}(\{\omega_i\}) > 0$ for each i . Determine all risk-neutral probability measures. Show that the European Call option is not attainable under Kim's model.
2. **(2020 exam, Q1)**
 - (a) Let S_t be the time t price of some stock and C_t^1 and C_t^2 the time t prices of European call options on that stock with maturity T and strike prices K_1 and K_2 where $K_1 < K_2$. We also consider a *capped* call option X on the same stock with same maturity T which has value

$$X_T = \begin{cases} K_2 - K_1, & \text{if } K_2 \leq S_T \\ S_T - K_1, & \text{if } K_1 \leq S_T \leq K_2 \\ 0, & \text{if } S_T \leq K_1. \end{cases}$$

Find the time t price X_t of the capped call option in terms of C_t^1 and C_t^2 .

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- (b) Consider a single-period model with one stock that has a current price of $S_0 = 1$ and a future price of either $S_1(\omega_1) = 1.3$ or $S_1(\omega_2) = 1.1$. The interest rate of the risk-free asset is r .
- i. Find all values for r such that there exists a risk neutral probability measure, and determine that measure in terms of r .
 - ii. Assume that $r = 0.4$ and that stock holders are getting paid a dividend of 0.1 at time point 1. Describe explicitly what an arbitrageur would have to do at time point 0 and time point 1 to gain riskless profit and calculate the profit in each state ω . Again assume that the arbitrageur does not have to pay any fees and that short selling is possible.